

Time: Three Hours

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer ALL Questions

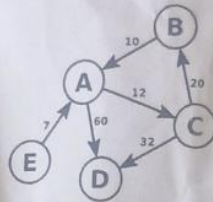
(10 X 10 = 100 Marks)

1. a) Obtain an upper bound for the following recurrence equation using recursive tree method and verify using Master's theorem [6+4]

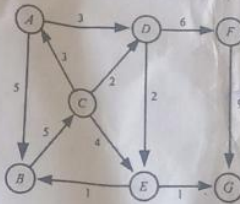
$$T(n) = \begin{cases} 1 & \text{when } n = 1 \\ 3T\left(\frac{n}{4}\right) + n^2 & \text{when } n > 1 \end{cases}$$

- b) Solve the following Knapsack problem which consists of divisible items using Greedy strategy. Knapsack capacity is 35. Find the items to be included into the knapsack bag to gain optimal profit, where the weights of items are 10, 20, 12 with corresponding profits 60, 100, 120.
2. Design a divide and conquer algorithm to find the maximum Subarray whose sum is maximum. Consider the array of numbers given by [2, -4, 1, 9, -6, 7, -3] to illustrate the algorithm. Trace the steps of the algorithm with the help of a recursive tree. Discuss its time complexity.
3. Design a dynamic programming algorithm for paranthesization of matrix chain which will require minimum number of scalar multiplications to find the product. Consider the chain of matrices given by A(2x3) B(3x2) C(2x4) and D(4x3) to illustrate the algorithm.
4. Solve the following instance of Knapsack problem using fixed length FIFO branch and bound strategy where we have 4 items and the knapsack bag has a capacity of 15. The profit values are {10, 12, 20, 18} and the corresponding weights are {2, 4, 5, 9}. Draw the annotated state space tree with cost and bound values. The steps for calculating bound values of various nodes should also be listed. Find the optimal solution and provide the list of items to be added into the knapsack bag.
5. Apply an algorithm that constructs longest prefix suffix (LPS) table for the pattern to find where it is plagiarized in a given text. List the algorithm and trace the steps for the pattern given by 'ABABAD' and the text given by 'ABCABABABAD'. Also return the position where the pattern is found in the text. Analyze its time complexity.

6. Find all pairs shortest path for the following weighted digraph using Floyd Warshall's Algorithm. Analyse its time complexity. Illustrate the tracing of the algorithm.



7. Apply Ford Fulkerson's algorithm on the following network flow graph to find the maximum flow from start vertex 'A' to sink vertex 'G'. In each iteration, the graph traversal performed to find the augmenting path, the flow value to be augmented and the residual flow network obtained after augmenting the flow should be drawn and described.

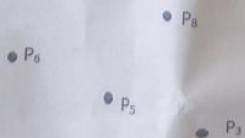


- 8.a) Design an algorithm for checking whether the two given line segments intersect or not. Trace and find if the line segment given by p_1q_1 and another line segment given by p_2q_2 intersect or not.
- $p_1(2,10)$ $q_1(7,2)$ and $p_2(4,15)$ $q_2(13,3)$
 - $p_1(6,4)$ $q_1(10,16)$ and $p_2(8,10)$ $q_2(13,6)$
 - $p_1(6,1)$ $q_1(7,3)$ and $p_2(8,4)$ $q_2(6,7)$
 - $p_1(2,3)$ $q_1(10,3)$ and $p_2(5,2)$ $q_2(7,3)$

OR

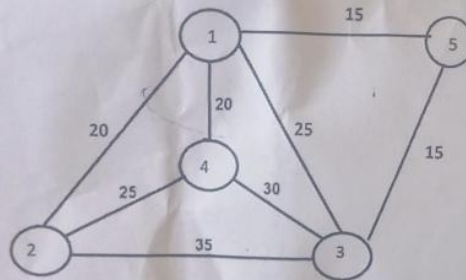
- 8.b) Consider the following set of points 'Q' in a XY plane. Construct a smallest polygon enclosing all the points. Design a $O(n \log n)$ time algorithm to find the same and trace it using the following points. Elaborate the step-by-step process of constructing the polygon.

$Q = \{P_1(10,3), P_2(6,1), P_3(8,4), P_4(2,2), P_5(3,5), P_6(1,6), P_7(7,3), P_8(6,7)\}$.



9. Apply Randomized quicksort algorithm by selecting the random pivot to trace the steps to sort in detail. Discuss how the average time complexity of randomized quicksort is better than the worst-case time complexity of deterministic quicksort. The elements are given below:
99, 34, 46, 22, 60, 23, 15, 14, 33, 20

- 10.(a) i. Prove that the approximation algorithm for solving Travelling Salesman problem is a polynomial time 2-approximation algorithm. [4]
ii. Design a polynomial time approximation algorithm for the travelling salesman problem. Illustrate the algorithm for the graph given below. Let 1 be the start vertex. [6]



OR

- 10.(b) Prove that "3-CNF (Conjunctive Normal Form) SAT" is NP complete. Use the satisfiability formula given below to prove the NP completeness. Illustrate the three steps for conversion of ϕ to ϕ''' . For two clauses, the detailed steps for converting ϕ' to ϕ''' are desired.

$$\phi = \overline{(x_4 \wedge ((x_1 \rightarrow x_2) \wedge (\overline{x_3} \Leftrightarrow x_1)))}$$

$$\Leftrightarrow \Leftrightarrow \vee / \wedge / \neg \Leftrightarrow \Leftrightarrow$$

School of Computer Science and Engineering

Winter Semester 2022-2023 Final Assessment Test

SLOT: G1

Programme Name & Branch : B.Tech CSE

Course Name & code: Design and Analysis of Algorithms BCSE204L

Class Number (s):

Exam Duration: 180 Min.

Maximum Marks:100

Q.No.	Question	Max Marks	CO	BL
1.	How does Karatsuba algorithm handle numbers with odd number of digits? Write the algorithm/program and show the stepwise execution of the same to multiply the two binary numbers – 10101 and 11. Analyse its time complexity.	10	CO1	BL3
2.	a) There are two knapsacks – one with a capacity of 8 and the weights and values of the items that can be used for filling it are (3,4,6,5) and (20,30,10,40) respectively and the other with a capacity of 20 and the weights and values of the items that can be used for filling it are (15,10,10) and (9,5,5) respectively. Validate the correctness of the statement “Greedy approach does not ensure an optimal solution for 0-1 Knapsack problem” using the given examples [5M] b) Two different character encoding schemes E1 and E2 encode the characters R, G, B, Y as (0, 011,111,11) and (0,11,101,100) respectively. The string “RGRBYRR” is encoded using both the methods, transmitted and decoded in the receiver side. Which of the two is Huffman encoding? Justify your answer. [5M]	10	CO1	BL5
3	Given below is the pseudocode to implement Dynamic Programming technique for matrix chain multiplication. Explain the reasons for assigning these specific initial and final values for the ‘i’ loop and ‘k’ loop and the ‘j’ variable.		CO2	BL2

```

for (i = 1 to n) m[i, i] = 0;
for (l = 2 to n)
{
    for (i = 1 to n - l + 1)
    {
        j = i + l - 1;
        m[i, j] = ∞;
        for (k = i to j - 1)
        {
            q = m[i, k] + m[k + 1, j] + p[i - 1] * p[k] * p[j];
            if (q < m[i, j])
            {
                m[i, j] = q;
                s[i, j] = k;
            }
        }
    }
}
return m and s;

```

Apply the same to fill the table given below and find the total number of scalar multiplications required to be done to multiply the following 5 matrices of ranks [4,10,3,12,20,7]. [5M+5M]

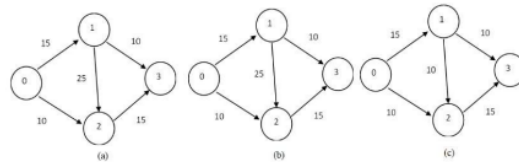
	0	1	2	3	4
0	0	120	264		
1		0	360	1320	
2			0	720	1140
3				0	1680
4					0

4	Write the code snippet / pseudocode for implementing backtracking technique to find the subset of elements from a given set, whose sum adds up to a given target value. Employ the same to find the first two subsets from the set [10,20,50,60,80] that sum up to the target 80.		CO2	BL3
5	Construct the Longest Prefix Suffix (LPS) table for the pattern "121231213". What is the length of the longest proper prefix and suffix that can be extracted from this pattern? What is the length of the proper prefix-proper suffix that can be extracted from the pattern "121231"? If the characters from indexes 0 to 6 of the pattern ("12123121") match with the text and the 7 th character does not match, which character in the pattern should the next comparison start from? Compare its time complexity with that of Naïve pattern-matching algorithm	10	CO3	BL3

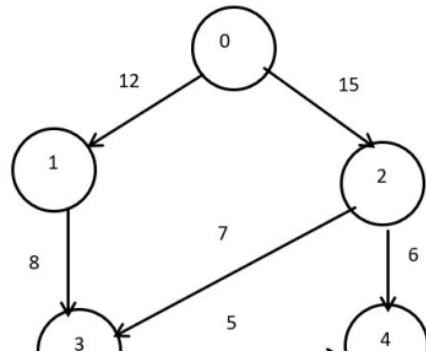
6.	“To find the maxflow using Ford Fulkerson algorithm, it is required to add the path flow along the reverse edges	10	CO4	BL4
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also in the residual graph, as there may be a need to send the flow in the reverse direction, sometimes”.
Following certain heuristics for the Ford Fulkerson algorithm, the first augmented path selected for the given three flow networks (a,b,c) are (0-1-3), (0-1-2-3) and (0-1-2-3) respectively. Proceed to calculate the maxflow for the 3 networks and determine which of the three scenarios help to justify the aforementioned statement.

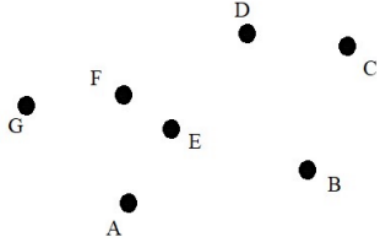
Consider source=0 and sink=3



7.	Apply Push-Relabel algorithm to find the maxflow for the graph given below. Source=0 and sink=4.	10	CO4	BL3
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8.	a) From the following set of line segments, find all the pairs that intersect, using a more efficient method than the Brute Force approach. [5M] $\{(2,8),(6,9)\},$ $\{(5,7),(10,7)\},$ $\{(3,5),(8,4)\},$ $\{(7,6),(12,12)\},$ $\{(9,10),(16,3)\},$ $\{(14,7),(17,9)\}$ b) Apply Graham's scan algorithm to find the convex hull of the following set of points. [5M]	10	CO5	BL3
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9.	Design a randomized algorithm for the hiring problem and perform probabilistic analysis to evaluate the computational complexity of the algorithm.	10	CO6	BL2
10.	Brief the various complexity classes and illustrate the conversion of SAT problem to 3-CNF-SAT problem.	10	CO7	BL1



Final Assessment Test – May 2024

Course: BCSE204L - Design and Analysis of Algorithms

Class NBR(s): 0765/0776/0817/0825/0829/0831/0839

/0874/0877/0884/0890/0901/0906/0912/0917/0923 Slot: A1+TA1

/0925/0927/0929/0931/0933/0939/6134

Time: Three Hours

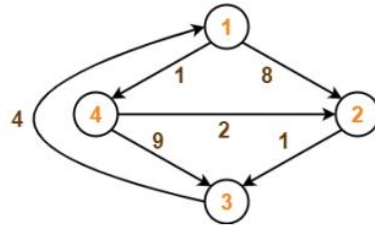
Max. Marks: 100

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- DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer any **TEN** Questions

(10 X 10 = 100 Marks)

1. Design an algorithm for sum of all elements of a matrix using 2D Array and analyze its time complexity. How to prove the correctness of matrix using Induction method. BL3 CO1
2. Outline important factors of Dynamic Programming and perform Matrix chain multiplication for {4, 10, 3, 12, 20, and 7}. The matrices have size 4 x 10, 10 x 3, 3 x 12, 12 x 20, 20 x 7. Compute $M[i, j]$, $0 \leq i, j \leq 5$. Consider $M[i, i] = 0$ for all i . BL2 CO2
3. a) Interpret steps of KMP_next array and the Knuth-Morris-Pratt (KMP) algorithm. Compute the KMP_next array of the Text as ABCCDDAEFG and pattern as CDD. [5] BL5 CO3
 b) Classify between KMP and Rabin-karp algorithm works with strength and weakness of Complexity. [5] BL2 CO3
4. Perform step-by-step process of find the shortest path distance between every pair of vertices using Floyd warshall algorithm, from the directed weighted graph. BL3 CO3



5. Relate Graham's Scan and Jarvis March Algorithm steps and how to choose the best practices for computational Convex Hull. Illustrate with an example. BL2 CO3
6. Narrate how Randomized Quick Sort algorithm differs from Normal Quick Sort. Write an algorithm and solve the randomized quick sort algorithm with an BL1 CO4

6. Narrate how Randomized Quick Sort algorithm differs from Normal Quick Sort. Write an algorithm and solve the randomized quick sort algorithm with an example.

BL1 CO4

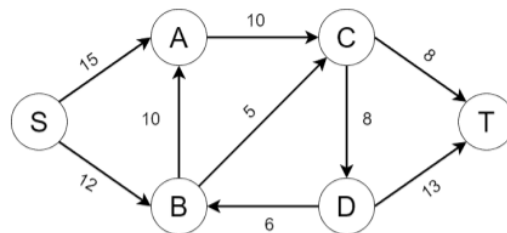
7. Compare and Contrast the following terms used in computational complexity such as N, NP and NP-complete problems. Discuss its features with real time examples.

BL2 CO5

Page 1 of 2

8. Write a Edmonds-Karp's algorithm, and compute the maximum flow in the network N and analyze its complexity.

BL3 CO3



9. Explain N-Queens algorithm and explain with 4 x 4 chessboard and number the rows and column of the chessboard 1 through 4 in a State space Tree using Backtracking (step-by-step checkerboard). Also derive Time complexity of N Queen using backtracking?

BL5 CO2

10. Define Greedy Knapsack? Find an optimal solution to the knapsack problem $n=7$, $M=15$, $(p_1, p_2, p_3, p_4, p_5, p_6, p_7) = (10, 5, 15, 7, 6, 18, 3)$, $(w_1, w_2, w_3, w_4, w_5, w_6, w_7) = (2, 3, 5, 7, 1, 4, 1)$.

BL1 CO2

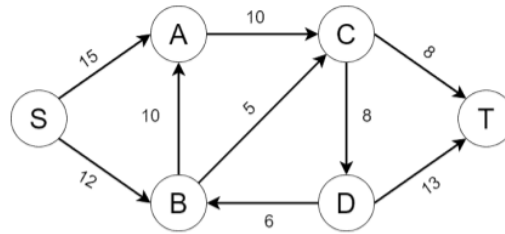
11. Solve the following tasks with their deadlines and profits. Schedule the tasks in such a way that they produce maximum profit after being executed

BL6 CO2

S. No.	1	2	3	4	5
Jobs	J1	J2	J3	J4	J5
Deadlines	2	2	1	3	4
Profits	20	60	40	100	80

8. Write a Edmonds-Karp's algorithm, and compute the maximum flow in the network N and analyze its complexity.

BL3 CO3



9. Explain N-Queens algorithm and explain with 4 x 4 chessboard and number the rows and column of the chessboard 1 through 4 in a State space Tree using Backtracking (step-by-step checkerboard). Also derive Time complexity of N Queen using backtracking?

BL5 CO2

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BL1 CO2

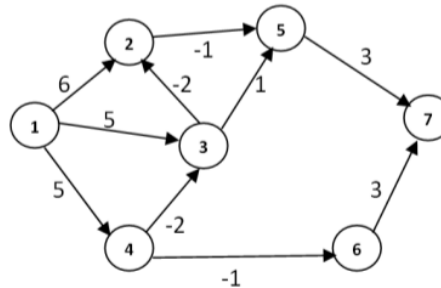
11. Solve the following tasks with their deadlines and profits. Schedule the tasks in such a way that they produce maximum profit after being executed

BL6 CO2

S. No.	1	2	3	4	5
Jobs	J1	J2	J3	J4	J5
Deadlines	2	2	1	3	4
Profits	20	60	40	100	80

12. Solve a Bellman ford single source shortest path computations and analyze its complexity. Consider starting vertex as "1" destination vertex as "7".

BL3 CO3



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VIT
Vellore Institute of Technology

Final Assessment Test – April 2023

Course: BCSE204L - Design and Analysis of Algorithms

Class NBR(s): 2864/2868/2869/2872/2874/2876/2878

/2879/2881/2882/2886/2889/2891/2895/2896/2897/
2899/2900/2902/2904/2905/2908/2910/4349/4351 Slot: G2+TG2

Time: Three Hours

Max. Marks: 100

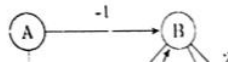
KEEPING MOBILE PHONE/SMART WATCH, EVEN IN "OFF" POSITION IS TREATED AS EXAM MALPRACTICE

General Instructions: Draw Diagram Wherever Necessary

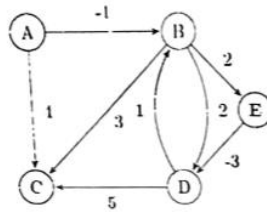
Answer ALL Questions

(10 X 10 = 100 Marks)

1. i. Write a non-recursive algorithm to find the sum of first 'n' odd numbers. [6]
Using the loop invariant property, prove the correctness of the algorithm.
- ii. Solve the following recurrence relations and find the time complexity [4]
for the following using master method.
a) $T(n) = 3T(n/3) + O(n^{1/2})$
b) $T(n) = 4T(n/2) + n^2$
2. State the fractional knapsack problem. Use the algorithm for fractional knapsack problem using greedy method and find an optimal solution for an instance with number of items $n = 7$, capacity of the sack $W = 23$, profit associated with the items $(p_1, p_2, \dots, p_7) = (29, 17, 20, 19, 6, 28, 42)$ and weight associated with each item $(w_1, w_2, \dots, w_7) = (7, 5, 6, 3, 9, 2, 5)$. Show the steps of your work.
3. Given a sequence of matrices $\langle 8, 5, 10, 20, 5 \rangle$. Using Dynamic programming find the minimum number of scalar multiplications needed and also write the optimal multiplication order.
4. Given the following number of jobs: $n = 4$, penalties $(p_1, p_2, p_3, p_4) = (5, 10, 6, 4)$ deadlines $(d_1, d_2, d_3, d_4) = (1, 2, 2, 1)$ and execution times $(t_1, t_2, t_3, t_4) = (1, 2, 1, 2)$. Find the actual cost and feasible jobs done using branch and bound technique.
5. For string matching, with the hash function Pattern (P) mod q , where $q = 13$, how many spurious hits does the Rabin-Karp matcher encounters in Text $T = 42637135421$ and $P = 54$ and also find the number of shifts to find the pattern. Write the pseudocode and also discuss about the time complexity of the same.
6. i. Elaborate the procedure to find the shortest path from source to all other vertices in the given graph using Bellman Ford algorithm and discuss how the dynamic programming strategy is applied for the same. [8]



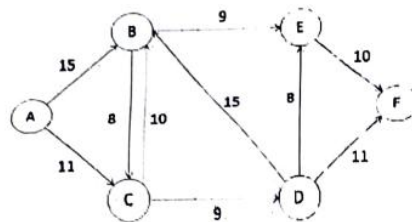
6. i. Elaborate the procedure to find the shortest path from source to all other vertices in the given graph using Bellman Ford algorithm and discuss how the dynamic programming strategy is applied for the same. [8]



- ii. State how Bellman Ford algorithm is used for detecting negative weight cycles. [2]

Page 1 of 2

7. Describe flow networks and apply the Ford – Fulkerson algorithm using augmented paths to find the possible maximum flow through the given flow network.

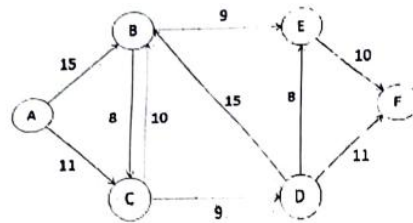


Discuss the step by step procedure along with its terminologies and pseudocode.

8. Given the following set of points 'P' in a plane. Construct the smallest polygon Q such that each and every point in 'P' is either on the boundary or inside the polygon. Use the Graham's Scan algorithm to wrap the points given below in counter clockwise direction to find the polygon.

(2,1)(3,3)(5,4)(-2,2)(-3,1)(-4,3)(-1,-2)(-3,-4)(2,-3)(4,-4)

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(2,1)(3,3)(5,4)(-2,2)(-3,1)(-4,3)(-1,-2)(-3,-4)(2,-3)(4,-4)

9. Write the pseudo code for solving hiring problem. Consider the cost to interview as c_i per candidate and the cost to hire the candidate is ch . Analyze the performance and prove that when the n candidates are presented in random order, then algorithm Hire-Assistant has a hiring cost of $O(ch \ln n)$.
10. Prove clique decision problem is NP-complete with the help of the following 3 CNF Sat.

$$(a \vee \bar{b} \vee \bar{c}) \wedge (\bar{a} \vee b \vee c) \wedge (a \vee b \vee c)$$

$\Leftrightarrow$

G/E/TX

Reg. No:



Final Assessment Test – May 2024

Course: BCSE204L - Design and Analysis of Algorithms

Class NBR(s): 0772/0796/0803/0822/0827/0834/

0876/0879/0887/0891/0902/0909/0914/0919/

0924/0926/0928/0930/0932/0934/0935/0937/

0941/6646

Slot: A2+TA2

Time: Three Hours

Max. Marks: 100

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- DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer any TEN Questions

(10 X 10 = 100 Marks)

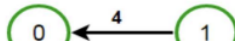
1. a) Solve the recurrence $T(n) = 4T(\sqrt{n}) + \log_2^5 n$ [4] BL3 CO1
b) Write an iterative algorithm to find the sum $\sum_{i=2}^n \frac{1}{i(i-1)}$, $n \in \mathbb{Z}$. Prove that your [6]
algorithm is correct.

2. You are given two DNA sequences, DNA1 and DNA2, consisting of characters from the alphabet {'A', 'C', 'G', 'T'}. Your task is to find the minimum number of single-character edits (insertions, deletions, or substitutions) required to transform DNA1 into DNA2. For example, given DNA1 = "AGTACG" and DNA2 = "GTCAGT", the minimum number of edits required is 4. One possible sequence of edits results in transforming DNA1 into DNA2 with 4 edits as follows: BL4 CO2

1. Substitute 'A' in DNA1 with 'G': "GGTACG"
2. Substitute 'G' in DNA1 with 'T': "GTTACG"
3. Delete 'A' from DNA1: "GTTCG"
4. Substitute 'C' in DNA1 with 'A': "GTTAG"

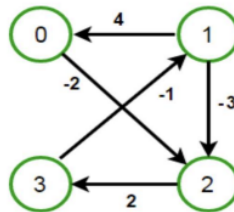
Develop a DP algorithm that will provide an optimized solution to determine the minimum edit distance between the two DNA sequences.

3. Assume that you have a text T and a pattern P . Write an algorithm to report the index of the first occurrence (if any) of P in T from the π -table of PT where PT is the concatenation of P and T . If P does not occur in T , then print -1. Trace the working of your algorithm with the help of a positive example. BL4 CO3
4. Given a directed weighted graph $G(V, E)$ with potentially negative weight edges, check whether the implementation of Bellman-Ford algorithm can find the shortest path between the nodes. How will your algorithm determine the possibility of negative cycles in the graph? Explain the reasoning behind the chosen approach. BL2 CO3



4. Given a directed weighted graph $G(V, E)$ with potentially negative weight edges, check whether the implementation of Bellman-Ford algorithm can find the shortest path between the nodes. How will your algorithm determine the possibility of negative cycles in the graph? Explain the reasoning behind the chosen approach.

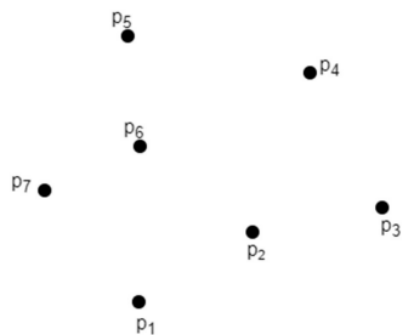
BL2 CO3



Page 1 of 3

5. Write the pseudocode for Jarvis-march algorithm and trace it for the points provided in the diagram below.

BL3 CO3

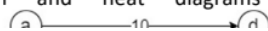


6. Illustrate the need of randomized algorithms. Demonstrate the working of randomized version of quicksort with an appropriate example.
7. Explain the clique decision problem. Show that it is NP complete by performing appropriate reductions.
8. a) Use Ford Fulkerson method (pseudocode/algorithm is NOT expected) to determine the maximum flow possible in the given flow network. Adequate explanation and neat diagrams are expected in each step.

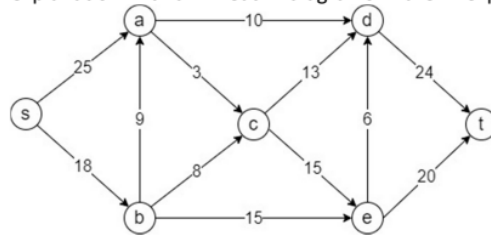
BL1 CO4

BL1 CO5

[7] BL3 CO3



8. a) Use Ford Fulkerson method (pseudocode/algorithm is NOT expected) to determine the maximum flow possible in the given flow network. Adequate explanation and neat diagrams are expected in each step. [7] BL3 CO3



- b) Identify an edge (u, v) in the network, in which decreasing its capacity by a positive integer k leads to a reduction of k in the maximum flow. Elucidate the mechanism by which this occurs. [3]
9. Define the subset sum problem. Solve it for the set $\{40, 30, 50, 10, 20\}$ with a target sum of 60. Use backtracking approach. BL2 CO2
10. You are attending a workshop with various sessions happening throughout the day. You'd like to attend as many sessions as possible without any conflicts. BL4 CO2

Session	Start Time	End Time
A	9:00	10:00
B	9:30	11:30
C	11:00	12:00
D	12:30	14:00
E	13:00	14:30
F	14:30	15:00

Find a greedy approach to find the maximum number of sessions you can attend without attending any part of two sessions simultaneously. Describe the algorithm and implement it to obtain the solution.

Page 2 of 3

11. Write the pseudocode for Rabin Karp algorithm and trace it for the text $T = \text{adbadbacfebgba}$ and pattern $P = \text{ba}$. The hash code for the characters is given in the following table. Assume the value of $d = 8$ and $q = 13$. Demonstration of rolling hashing function is mandatory. Coding scheme: a-1, b-2,.... BL3 CO3
12. In the field of robotics, path planning algorithms are crucial for robots to navigate BL3 CO3

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12. In the field of robotics, path planning algorithms are crucial for robots to navigate their environment efficiently. Imagine a robot vacuum cleaner that needs to clean the edges of a room. How can the Graham Scan algorithm be used to determine the most efficient cleaning path along the room's perimeter. The coordinates given for this problem are:
 $\{(-3 -4), (2 -3), (4 3), (-4 2), (0 5), (2 -3), (-1 4)\}$.

BL3 CO3

BL3 CO3

↔↔↔G/E/TX↔↔↔

U/D/TY



VIT
Vellore Institute of Technology

Final Assessment Test – April 2025

Course: BCSE204L - Design and Analysis of Algorithms

Class NBR(s): 1459/1472/1489/1496/1504/1510/1518/
1525/1530/1534/1537/1543/1547/1555/
1561/1567/1578/1586/1594/1600/1608

Reg. No: 23BCE0990

Slot: A1+TA1

Time: Three Hours

Max. Marks: 100

- > KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
> DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer **ALL** Questions
(10 X 10 = 100 Marks)

1. The Karatsuba algorithm is a divide-and-conquer approach for fast integer multiplication. Break down the Karatsuba algorithm into its constituent stages: problem decomposition, recursive subproblems, and recombination. Derive and analyze its time complexity, comparing it with the standard multiplication method. Illustrate the algorithm for the multiplication of two integers 21 and 143.
2. Construct the optimal Huffman code for a given set of character frequencies derived from the first 8 prime numbers.

Character	a	b	c	d	e	f	g	h
frequency	2	3	5	7	11	13	17	19

Derive the time and space complexity of the Huffman code. Define the prefix free property and the need for prefix free code in Huffman coding. Show that Huffman coding takes less space as compared to fixed size code.

3. A logistics company wants to optimize the delivery route for a salesperson who must visit four cities exactly once before returning to the starting city. The company has recorded the travel distances between cities in the given matrix in Table 2.

As a software engineer, develop an efficient route-planning algorithm using

3. A logistics company wants to optimize the delivery route for a salesperson who must visit four cities exactly once before returning to the starting city. The company has recorded the travel distances between cities in the given matrix in Table 2.

As a software engineer, develop an efficient route-planning algorithm using Dynamic Programming (DP).

- Formulate the Traveling Salesman Problem (TSP) using the DP approach and explain its optimal substructure.
- Using the Held-Karp Algorithm (Dynamic Programming), compute the minimum travel cost for the given distance matrix and show step-by-step calculations.

- c) Compare the time and space complexity of the dynamic programming approach with the brute-force approach.

	A	B	C	D
A	0	4	1	9
B	3	0	6	4
C	4	1	0	2
D	6	5	-4	0

Table 2: Distance matrix between cities.

A delivery company wants to optimize the selection of delivery packages to maximize profit while ensuring that the total weight of selected packages does not exceed the weight limit of their delivery van. Each package has a specific weight and value associated with it.

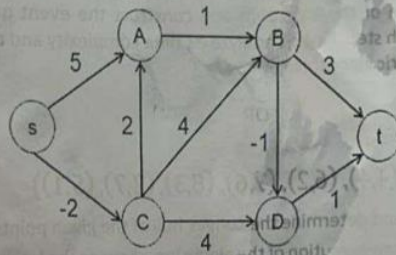
As an algorithm specialist, apply the **FIFO Branch & Bound** approach to solve the **0-1 Knapsack Problem** with the following items:

Package	Weight (kg)	Value (Rs.)
1	2	40
2	3	50
3	4	70
4	5	80

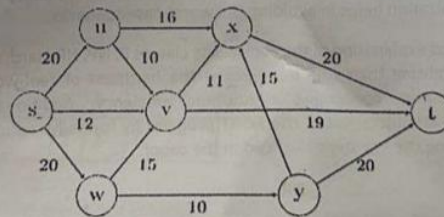
Contrast the Backtracking and Branch & Bound paradigms in solving the 0-1 Knapsack Problem, highlighting their differences in terms of search strategy, pruning mechanism, efficiency, and applicability.

5. As a bioinformatics researcher, you are tasked with developing an efficient algorithm to identify specific genetic markers in DNA sequences. You need to search for the genetic marker pattern $P = \text{"ACGTACGT"}$ within the target DNA sequence $T = \text{"GCACGTACGTGACGTACGTA"}$. Apply the KMP and Rabin-Karp algorithm for finding all occurrences of the pattern P in the given text T . Trace the execution of your chosen algorithm when searching for pattern P in sequence T , highlighting key comparisons and shifts. For Rabin Karp, consider $A=1, C=2, G=3, T=4$, base 4, and mod 101 and show the number of spurious hits. Compare the efficiency of your solution by calculating the total number of character comparisons performed by both algorithms for this specific example. Identify the algorithm that runs faster as compared to the other.

Apply the Bellman Ford algorithm to compute the all-pairs shortest paths for the given weighted graph. Show the step-by-step execution of the algorithm. Discuss the strength and weakness of the algorithm.



7. A logistics company is planning the most efficient way to transport goods from a central warehouse (source s) to a major distribution center (destination t) through a network of cities: u, v, w, x, y . Each road connecting these cities has a specific maximum carrying capacity, representing the amount of goods that can be transported at a time, represented by the graph given below.



Currently, no goods are flowing through the network. The company wants to maximize the shipment flow using the Ford-Fulkerson algorithm while following these conditions:

- A. In the first step, the route with the **longest augmenting path** (largest number of cities) should be chosen for transportation.
- B. In the second step, the route with the **highest residual capacity** should be prioritized.

In your answer, clearly show (i) the original graph with all capacities filled up, (ii) the augmenting path chosen in the first step (write this as a sequence of vertices, do not mark on the graph), (iii) the flow network after the first augmentation, (iv) the corresponding residual graph after the first augmentation, (v) the augmenting path chosen in the second step, (vi) the flow network after the second augmentation, (vii) the corresponding residual graph after the second augmentation. viii) Also write the time and space complexity analysis of the algorithm.

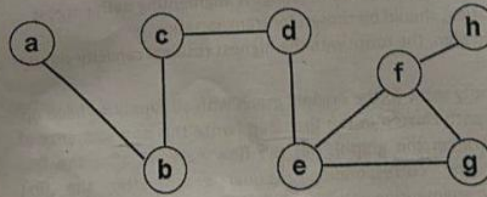
- i) Given the set of points $P = \{(1,2), (3,5), (4,4), (6,2), (7,6), (8,3)\}$ and line segments formed by connecting $(1,2) \rightarrow (4,4)$, $(3,5) \rightarrow (7,6)$, $(6,2) \rightarrow (8,3)$, and $(3,5) \rightarrow (6,2)$. Apply the sweep line algorithm to detect intersections among these segments. Demonstrate the step-by-step execution of the algorithm and construct the event queue and status structure at each step. Finally, analyze its time complexity and discuss its significance in geometric algorithms.

OR

- b) Given the set of points:
 $P = \{(1,2), (3,5), (4,4), (6,2), (7,6), (8,3), (2,7), (5,1)\}$
 Apply **Graham's Scan** and determine the convex hull of the given points. Illustrate the step-by-step execution of the algorithm, showing the intermediate hull formations. Compare computational efficiency of **Graham's Scan** and **Jarvis' March** in terms of **time complexity** and discuss the conditions under which one method may be preferred over the other.
9. Apply the **Randomized QuickSort** algorithm to sort the array $[9, 4, 7, 2, 6, 3, 8]$. Show the step-by-step execution, including the selection of random pivots, partitioning, and recursive calls. Analyze its expected time complexity and discuss how randomization helps in avoiding the worst-case scenario.
10. a) Provide a summarized explanation of the complexity classes P, NP, NP-Hard, and NP-Complete, highlighting their role in assessing the hardness of real-world computational problems concerning algorithmic efficiency. Additionally, demonstrate the NP-completeness of the 3-SAT problem by formally reducing SAT to 3-SAT, outlining the key steps involved in the proof.

OR

10. b) Provide a summarized explanation of the Vertex Cover Problem, its computational complexity, and its significance in real-world applications. Discuss why it is classified as NP-complete and demonstrate the proof that Vertex Cover has a 2-approximation algorithm, outlining the key steps involved in the approximation technique. Obtain a vertex cover for the following graph.



⇔⇔⇔ U/D/TY ⇔⇔⇔

U/D/TZ

Reg. No:



Final Assessment Test – April 2026

Course: BCSE204L - Design and Analysis of Algorithms

Class NBR(s): 3559/3561/3585

Slot: A1+TA1

Time: Three Hours

Max. Marks: 100

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
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COs	CO Statements
CO1	Apply the mathematical tools to analyze and derive the running time of the algorithms
CO2	Demonstrate the major algorithm design paradigms.
CO3	Explain major graph algorithms, string matching and geometric algorithms along with their analysis.
CO4	Articulating Randomized Algorithms.
CO5	Explain the hardness of real-world problems with respect to algorithmic efficiency and learning to cope with it.

BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

Answer ALL Questions

(10 X 10 = 100 Marks)

1. a) Design an algorithm for the Fractional Knapsack problem using the greedy [5] CO1 BL3 approach. Given two lists of equal length:
 Profits: [5, 10, 15, 7, 8, 9]
 Weights: [1, 3, 5, 4, 1, 3]
 Determine the maximum profit that can be stored in a Knapsack of capacity $W = 15$ Kg. You must use the Fractional Knapsack approach. Show the calculation of the profit/weight ratio for each item and state the order in which items are selected. Why is the Fractional Knapsack problem solved using a Greedy approach, while 0/1 Knapsack cannot be?
- b) Obtain an upper bound for the following recurrence equation using [5] recursive tree method and verify using Master's theorem

$$T(n) = \begin{cases} 1 & \text{when } n = 1 \\ 3T\left(\frac{n}{4}\right) + n^2 & \text{when } n > 1 \end{cases}$$

2. Design the pseudocode for Divide and Conquer Maximum Subarray and Apply algorithm to find the contiguous subarray with the largest sum in the following array:

CO1 BL3

A = [-2, 5, -1, 3, -4, 7, -2, 1]

Show the left sum, right sum, and cross sum for each recursive division. Clearly state the starting and ending indices of the maximum subarray and its sum.

- 3.(a) Consider the N-Queens problem for N = 6. Construct the state-space tree up to level 3. Clearly indicate promising and non-promising nodes. Show the complete backtracking process to obtain one valid solution. Modify the traditional backtracking approach to reduce unnecessary safety checks.

CO2 BL4

Analyze the worst-case time complexity and compare it with brute-force permutation generation

OR

- 3.(b) You are attending a workshop with various sessions happening throughout the day.

CO2 BL4

You'd like to attend as many sessions as possible without any conflicts.

Session	Start Time	End Time
A	9:00	10:00
B	9:30	11:30
C	11:00	12:00
D	12:30	14:00
E	13:00	14:30
F	14:30	15:00

Find a greedy approach to find the maximum number of sessions you can attend without attending any part of two sessions simultaneously. Describe the algorithm and implement it to obtain the solution

4. Solve the Travelling Salesman Problem for the following graph using Dynamic Programming approach :

CO2 BL3

Cost Matrix (Distance between cities):

	A	B	C	D
A	∞	20	42	35
B	20	∞	30	34
C	42	30	∞	12
D	35	34	12	∞

Starting city: A

Show all calculations in detail. Determine the optimal tour and its cost. List the order of cities visited. Trace back to reconstruct the optimal path. Explain why Dynamic Programming becomes increasingly more efficient than Brute Force as the number of cities increases.

5. Given:

CO3 BL4

- Text T: "14159265358979323846" (first 20 digits of π)
- Pattern P: "59265"

Assume: Base $d=10$ and Prime $q=13$ (Hint: $H = (\text{digit}_0 * d^{(m-1)} + \text{digit}_1 * d^{(m-2)} + \dots + \text{digit}_{(m-1)}) \bmod q$)

Compute the initial hash values for the pattern and first window. Show rolling hash computation for next three shifts. Identify spurious hits (if any). Explain why Rabin-Karp performs well for multiple pattern matching. Analyse average and worst-case time complexity.

6. Given weighted directed graph:

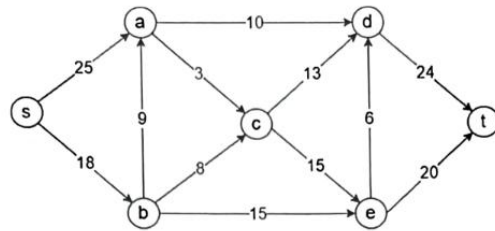
CO3 BL3

- B — A: Weight 10
- A — C : Weight 12
- C — B: Weight 20
- C — D: Weight 32
- A — D: Weight 60
- E — A: Weight 7

Find all pairs shortest path for the given weighted digraph using Floyd Warshall's Algorithm. Analyse its time complexity. Illustrate the tracing of the algorithm. Detect whether a negative cycle exists using diagonal entries.

7. Apply Ford Fulkerson's algorithm on the following network flow graph to find the maximum flow from start vertex 's' to sink vertex 't'. In each iteration, the graph traversal performed to find the augmenting path, the flow value to be augmented and the residual flow network obtained after augmenting the flow should be drawn and described.

CO3 BL3



8. Consider the following set of points 'p' in a XY plane. Construct a smallest polygon enclosing all the points. Design a $O(n \log n)$ time algorithm to find the same and trace it using the following points. Elaborate the step-by-step process of constructing the polygon.

CO3 BL3

P1(10,3), P2(6,1), P3(8,4), P4(2,2), P5(3,5), P6(1,6), P7(7,3), P8(6,7)

9. Apply Randomized quicksort algorithm by selecting the random pivot to trace the steps to sort in detail. Discuss how the average time complexity of randomized quicksort is better than the worst-case time complexity of deterministic quicksort. The elements are given below:

CO4 BL3

99, 34, 46, 22, 60, 23, 15, 14, 33, 20

- 10.a) Provide a summarized explanation of the complexity classes P, NP, NP-Hard, and NP-Complete, highlighting their role in assessing the hardness of real-world computational problems concerning algorithmic efficiency.

CO5 BL5

OR

- 10.b) Prove that the Clique Decision Problem is NP-Complete using 3-CNF SA

CO5 BL5

$$(a \vee \bar{b} \vee \bar{c}) \wedge (\bar{a} \vee b \vee c) \wedge (a \vee b \vee c)$$

$$\Leftrightarrow U/D/TZ \Leftrightarrow$$